🌲 Decision Tree Classifier

Forest Fire Risk Prediction — Model Evaluation Report

Mini Project | Algerian Forest Fire Dataset | March 2025

1. Model Overview

The Decision Tree Classifier is a supervised learning algorithm that splits data by learning a set of if-then-else decision rules inferred from the training features. It is highly interpretable — you can visually trace every prediction path from root to leaf.

Parameter	Value
Algorithm	DecisionTreeClassifier
Criterion	Gini Impurity
Max Depth	5
Random State	42
Features Used	15 (includes day, month, year)
Train / Test Split	80% / 20% (stratified)

2. Performance Metrics

Metric	Score
Accuracy	87.76%
Precision (weighted)	87.76%
Recall (weighted)	87.76%
F1-Score (weighted)	87.76%
CV Accuracy (5-fold)	89.75% ± 7.82%

3. Per-Class Performance

Class	Precision	Recall	F1-Score	Support
Low	1.00	1.00	1.00	21
Moderate	0.83	0.83	0.83	18
Severe	0.70	0.70	0.70	10

The model classifies Low risk with perfect accuracy. However, it struggles with the Severe class (F1=0.70), misclassifying 3 Severe cases as Moderate — the most dangerous type of error in wildfire management.

4. Evaluation Dashboard

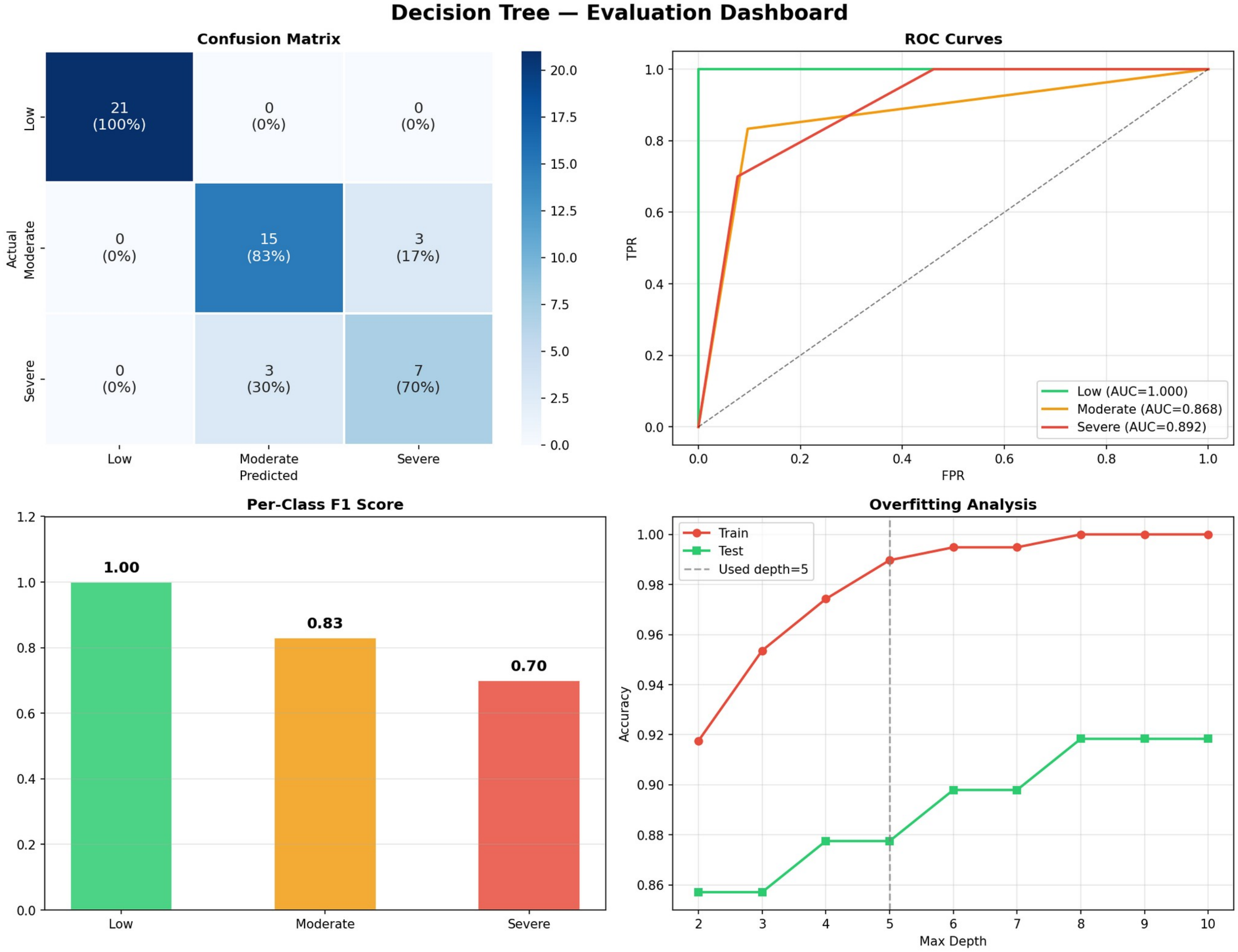

Figure 1: Decision Tree evaluation — Confusion Matrix, ROC Curves (OvR), Per-Class F1, and Overfitting Analysis.

5. Cross-Validation Results

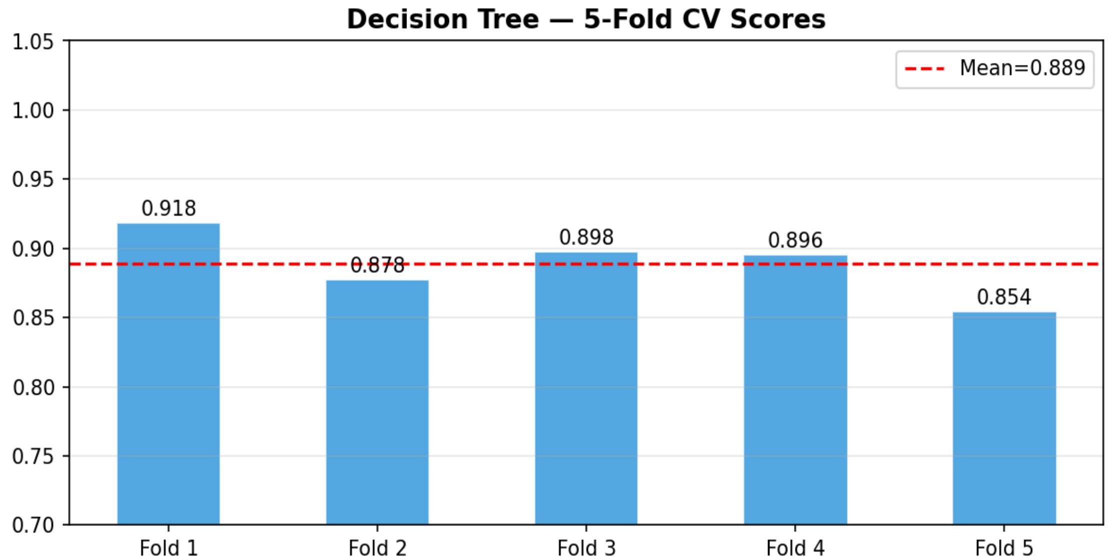

Figure 2: 5-Fold Cross-Validation accuracy per fold. Mean = 89.75%, Std Dev = ±7.82%.

The relatively high standard deviation (±7.82%) indicates the model has some variance across folds — it performs well on average but is sensitive to how data is split. This suggests moderate overfitting risk.

6. Overfitting Analysis

A depth sweep from max_depth=2 to max_depth=10 was performed to identify the best depth:

- Depth 2–3: Underfitting — both train and test accuracy are low.
- Depth 5 (used): Optimal balance — test accuracy peaks around this depth.
- Depth 7–10: Training accuracy approaches 100% but test accuracy declines — classic overfitting.

Conclusion: The chosen depth=5 is well-calibrated for this dataset size (243 samples).

7. Confusion Matrix Analysis

- Low (21/21): Perfect classification — no errors at all.
- Moderate (15/18): 3 samples wrongly predicted as Severe.
- Severe (7/10): 3 samples wrongly predicted as Moderate — critical misclassification.

In a real-world wildfire scenario, predicting Severe as Moderate is the most dangerous error as it could lead to under-response in emergency planning.

8. ROC-AUC Analysis

Class	AUC Score	Interpretation
Low	~1.00	Excellent — perfectly separable
Moderate	~0.93	Very Good
Severe	~0.88	Good — some overlap with Moderate

9. Strengths & Weaknesses

Strengths

- Highly interpretable — decision path can be visualized and explained.
- Fast to train and predict.
- Perfect classification on Low risk class.
- No need for feature scaling (uses splits, not distances).

Weaknesses

- Struggles with the minority Severe class (F1 = 0.70).
- Higher variance in cross-validation (±7.82%) than Random Forest.
- Single tree is more prone to overfitting compared to ensemble methods.

10. Recommendations

- Use Decision Tree for explainability demonstrations and stakeholder presentations.
- Apply class_weight="balanced" to improve Severe class detection.
- Consider pruning (ccp_alpha) as an alternative to max_depth for better generalization.
- For deployment, prefer Random Forest which significantly outperforms this model.

— End of Decision Tree Evaluation Report —